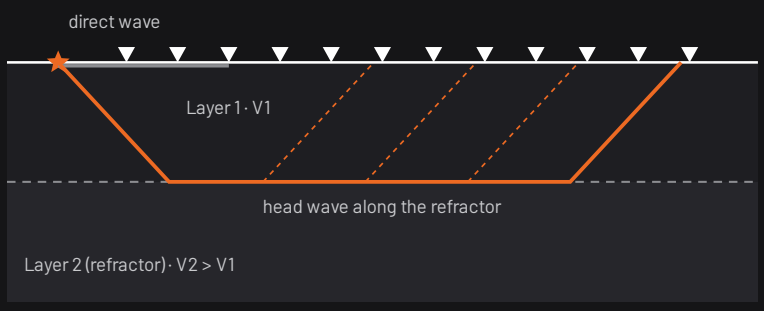
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FIELD GUIDE

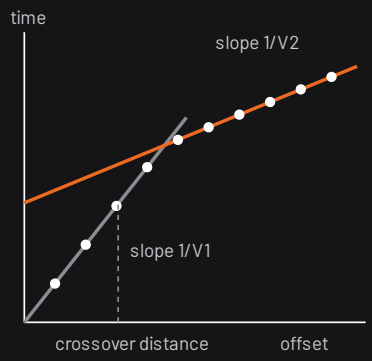
Seismic Refraction Field Guide

SET UP · RECORD · ANALYSE

Equipment, line layout, recording P and SH shots, and the analysis in Refraction Studio, step by step, for your first survey.



CROSS-SECTION



FIRST-ARRIVAL TIMES

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QUAKELOGIC

1. Before you start

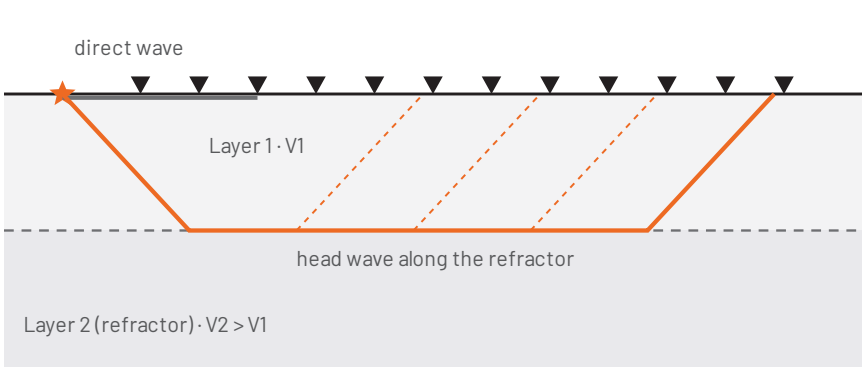
This guide takes you from the equipment case to a finished seismic refraction report. It assumes no previous experience. Read it once in the office, then take the checklists (section 11) to the field.

WHAT SEISMIC REFRACTION MEASURES

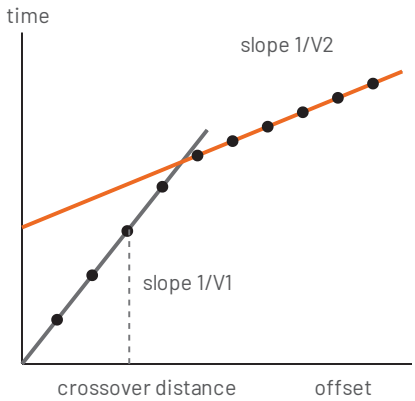
When a hammer hits the ground, waves travel down until they reach a faster layer, such as dense gravel, the water table (for P waves) or bedrock. There they bend and run along the top of the faster layer, sending energy back up to the surface. Far from the shot, these refracted waves overtake the waves travelling directly along the surface.

A line of geophones records the **first arrival** (the first break) at every position. Refraction Studio picks these times and turns them into the velocity of each layer and the depth to each refractor. It can also compute a gradational velocity section (tomography).

- **P waves** (compressional) give the P-wave velocity (Vp). Use them for depth to bedrock, rippability, the water table and layer thickness.
- **SH waves** (horizontal shear) give the shear-wave velocity (Vs). Use them for stiffness, Vs30 site class and, with P waves, Poisson's ratio and the elastic moduli.



CROSS-SECTION



FIRST-ARRIVAL TIMES

Direct and refracted waves, and the travel-time curve they produce

You need three things right in the field: well-planted geophones at measured positions, a sharp trigger at the moment of every hit, and shots at both ends of the line. Everything else can be checked and corrected in Refraction Studio.

Know the limits. Refraction assumes that velocity increases with depth. A slower layer under a faster one, or a very thin layer, can be missed. Tomography and a second line at a different spacing help; borehole logs remain the best check.

THE EQUIPMENT LIST

Item	What to bring	Notes
Seismograph	Recording unit or master unit, its cables and the laptop with the recording software	24 channels or more; 48 gives better tomography
Line cable	Cable with a take-out (or digitizer shell) for every channel	The take-out spacing limits the geophone spacing
Geophones, P line	One **vertical** geophone per channel, 10–14 Hz or 4.5 Hz, with spikes	Plus two spares
Geophones, SH line	One **horizontal** geophone per channel, 10–14 Hz or 4.5 Hz, with spikes and a bubble level	Plus two spares
Trigger	Trigger switch on the hammer handle, or a trigger geophone next to the plate	Starts every recording
Reference geophone (optional)	An extra geophone next to the shot, on a spare channel	Lets Refraction Studio correct trigger timing errors
P source	Sledgehammer, **5–10 kg (10–20 lb)**, and a nylon or aluminium strike plate	A weight drop for long lines
SH source	Shear beam: a hardwood beam, about 2–3 m (7–10 ft) long, with steel end plates	Held down by a vehicle wheel or about 200 kg (440 lb) of weights
Measuring tape	100 m (330 ft) tape, plus pegs	Lay it along the whole line
Markers	Survey flags, spray paint or pegs	Mark every geophone and shot position
Elevations	Level and staff, total station or GNSS/GPS receiver	Needed whenever the ground is not flat
Power	Charged batteries for the seismograph and laptop	Check them the day before
Field sheet	Printed field sheet (section 11) and a pencil	Write down every file
Safety	Ear defenders, safety glasses, gloves, safety boots, high-visibility vest	The hammer operator needs all of them

CHECK EVERYTHING IN THE OFFICE

1. Charge every battery and the laptop.
2. Connect the seismograph to the laptop and start the recording software. Check that every channel answers and that the trigger works (tap the trigger switch or the trigger geophone and watch for a recording to start).
3. Check that the software saves records as **SEG-2** (`.sg2`, `.seg2`, `.dat`) or **SEG-Y** (`.sgy`, `.segy`); Refraction Studio reads both.
4. Install Refraction Studio on the computer you will use for the analysis (section 10). It can be the same laptop.

2. Plan the line

PICK THE PLACE

- Choose a **straight line** across the feature you want to see, for example across a buried valley rather than along it.

- Prefer open ground. Slopes are fine if you measure the elevations (section 8).
- Keep away from **noise**: engines, generators, pumps, busy roads and people walking near the geophones. If you cannot avoid traffic, record between vehicles.
- Avoid lines that run along buried pipes, walls or foundations; they carry waves of their own.

HOW DEEP CAN YOU SEE?

The depth you can see depends on the **length of the geophone line** (the spread). A good rule of thumb: the spread should be **three to five times the depth you need**. Off-end shots (see below) add depth without adding geophones.

Wanted depth	Geophone spacing	Spread (24 channels)	Spread (48 channels)	Source
3–8 m (10–25 ft)	1–2 m (3–6.5 ft)	23–46 m (75–150 ft)	47–94 m (155–310 ft)	5 kg (10 lb) hammer
8–20 m (25–65 ft)	3 m (10 ft)	69 m (225 ft)	141 m (460 ft)	8–10 kg (16–20 lb) hammer, several hits per shot
20–40 m (65–130 ft)	5 m (16 ft)	115 m (380 ft)	235 m (770 ft)	Weight drop, or many stacked hammer hits

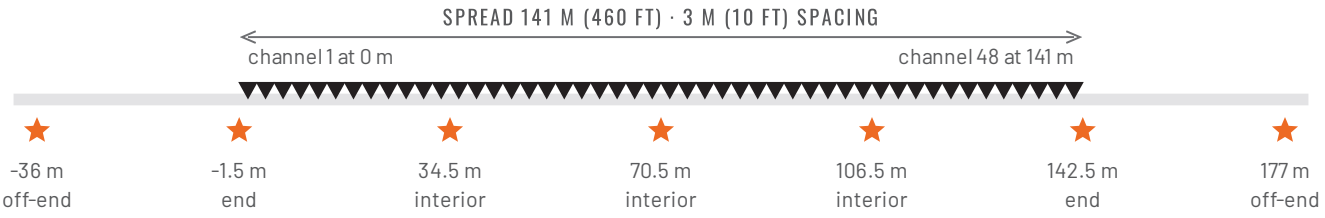
- **Closer spacing** shows thin near-surface layers and short changes along the line.
- **Vs30** needs the SH model to reach 30 m (100 ft). That usually needs a long SH spread with off-end shots.

WHERE THE SHOTS GO

Record shots at several positions along the same line. Each shot gives one travel-time curve; together they give the reverse coverage that the layered methods need and the ray coverage that tomography needs.

Shot	Position	Why
End shots	Half a spacing beyond the first and last geophone	Forward and reverse times for the delay-time and plus–minus methods
Off-end shots	Beyond both ends, at about one-quarter of the spread length	Deeper refractor coverage under the end geophones
Interior shots	Every 6 to 12 geophones along the line	Detail along the line and ray coverage for tomography

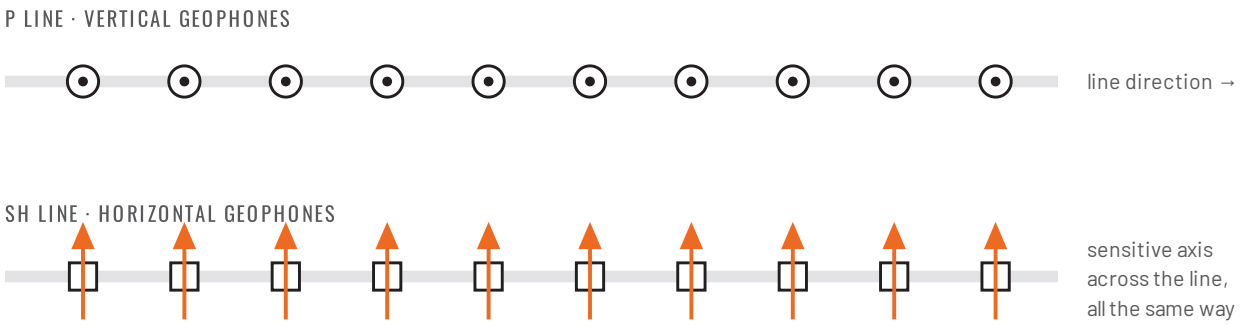
Minimum: both end shots and one centre shot. **Recommended:** seven shots or more, as in the sample project: two off-end, two end and three interior shots on a 48-channel line at 3 m spacing.



Shot layout of the sample project: two off-end, two end and three interior shots

3. Lay out the line

1. **Stretch the measuring tape** along the line and fix both ends. Decide which end is **channel 1** and write it on the field sheet.
2. **Mark the positions.** Put a flag or paint mark at every geophone position (for example 0, 3, 6 … 141 m) and at every shot position.
3. **Lay out the line cable** along the tape, with channel 1 at the first mark. Keep it flat on the ground; a cable moving in the wind adds noise.
4. **Plant the geophones.** Scrape away grass and loose topsoil, then push each geophone in by hand until its base sits on the ground. Tap the spike in if needed, but never hit the geophone case.
5. **P line: vertical geophones upright,** within the tilt limit on the geophone's data sheet (typically 10–20°). A tilted or loose geophone gives a weak, noisy trace.
6. **SH line: horizontal geophones level and turned the same way,** with their sensitive axis **across the line** (perpendicular to it) and parallel to the shear beam. Use the bubble level. A geophone turned the wrong way records the wrong polarity.
7. **Connect each geophone** to its take-out. Keep the connectors clean and dry.



Plan view: geophone orientation on P and SH lines

On pavement or concrete you cannot use spikes. Use geophone base plates or tripod stands, or sit the geophones on small sandbags. Each geophone must still be firmly in contact with the ground.

4. Set up the seismograph

1. **Connect** the line cable, the trigger and the power to the seismograph, then the seismograph to the laptop.
2. **Start the recording software** and check that every channel is found. A missing channel means a loose connector or a faulty take-out.
3. **Choose a new folder** for each site or line. Every record is saved there.
4. **Enter the positions** if the software allows it: the X position of every channel, and the shot position before every shot. Elevations go in as Z. Refraction Studio reads them from SEG-2 and SEG-Y headers; otherwise you enter the layout in the Geometry page.

RECORDING SETTINGS

Setting	P line	SH line	Why
Sample interval	0.125–0.25 ms (4,000–8,000 samples per second)	0.25–0.5 ms	First breaks must be timed to a fraction of a millisecond
Record length	0.25–0.5 s	0.5–1 s	Long enough for the first arrival at the farthest geophone, plus margin
Delay / pre-trigger	0, or a pre-trigger saved in the file	0, or a pre-trigger saved in the file	Refraction Studio applies the delay written in the file header
Filters	Off (or a low-cut below 10 Hz)	Off	Filters in the field shift first breaks; filter later in Refraction Studio
Gain	The same on every channel	The same on every channel	Fixed gains keep amplitudes comparable
Stacking	3–10 hits per shot point	5–10 hits per side	More hits raise the signal above the noise

Record length rule: the farthest offset divided by the slowest velocity, plus 50 %. For a 180 m (590 ft) offset and a slowest P velocity of 400 m/s, that is 0.45 s plus margin: record 0.5 s.

NOISE CHECK

Before the first shot, watch the noise display with nobody moving. Every channel should show a small, steady signal. A flat channel is not connected; a channel with large jumps is loose, in the wind or next to someone walking.

5. Record the P shots

1. **Place the strike plate** on the shot mark and press it firmly into the ground. Put the trigger geophone right next to the plate, or check the hammer switch.
2. **Optional reference geophone:** plant a geophone about 0.5–1 m (2–3 ft) from the plate on a spare channel, and note the channel number. Refraction Studio uses it to correct trigger timing errors (channel role **Reference** in Geometry).
3. Make sure nobody is walking near the line. The hammer operator wears ear defenders, safety glasses and gloves; everyone else stands back.
4. **Arm the recording** and hit the centre of the plate squarely with a full swing.
5. **Repeat and stack** 3–10 hits at the same point (in the seismograph, or keep the records separate and stack them in Refraction Studio at import).
6. **Check the record** (below), then write the file name, shot position and notes on the field sheet.
7. **Move to the next shot point** and update the shot position.

WHAT A GOOD RECORD LOOKS LIKE

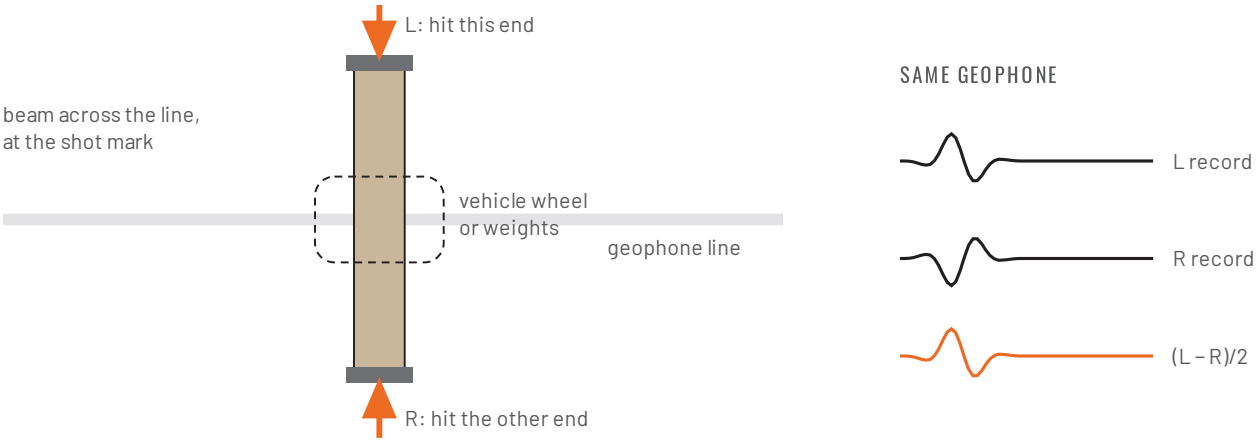
- **Good:** the first breaks form a clear line across the record, earliest next to the shot and later at every farther geophone, often with a change of slope where the refracted wave overtakes the direct wave.
- **Dead channel:** one flat trace. Check the geophone and its connector.
- **Noisy channel:** one trace full of jitter. Re-plant the geophone and check for wind.

- **Clipped:** flat-topped waveforms next to the shot. That is normal on the nearest one or two geophones; on more, hit more softly or lower the gain.
- **Weak far traces:** first breaks are lost in the noise at the far end. Stack more hits, use a heavier source or add off-end shots.

6. Record the SH shots

SH waves are generated by striking a beam sideways. Hitting it from the left and then from the right reverses the shear wave but not the P wave, which lets Refraction Studio separate them.

1. **Lay the shear beam** on the shot mark, **across the line** (perpendicular to it), parallel to the sensitive axis of the horizontal geophones.
2. **Load the beam** so it cannot slide: park a vehicle wheel on it, or stack about 200 kg (440 lb) of weights. Good contact with the ground matters more than the weight of the hammer.
3. **Hit one end** of the beam horizontally, along its length, and record. Stack 5–10 hits. Name the file with **L** (for example `shot_003_L.sg2`).
4. **Hit the other end** the same number of times and record a separate file named with **R** (`shot_003_R.sg2`). Do not stack left and right hits together.
5. **Check the pair:** the shear-wave arrivals on the L and R records must have opposite polarity. If they look the same, the geophones or the hits are aligned wrongly.
6. Repeat at every SH shot position, using the same positions as the P line when you want Poisson's ratio and the elastic moduli.



SH source: a loaded shear beam struck from both ends gives records of opposite polarity

In Refraction Studio, import the L and R records of a line together and choose **Opposite SH impacts** under *Records at the same position*. Each pair is combined as (left – right)/2: shear energy adds and P energy cancels.

7. Recording log and file names

- Use one folder per line and wave type (for example `Line1_P` and `Line1_SH`).
- Name files by shot number, with L/R for SH: `shot_001.sg2`, `shot_001_L.sg2`.

- Write every file on the field sheet with its shot position, the number of hits and anything unusual (traffic, wind, a re-planted geophone).

8. Sloping ground and elevations

Refraction depths are measured from the ground surface, so elevations matter whenever the ground is not flat.

1. Lay the tape **along the ground** and mark positions along the tape as usual.
2. Measure the **elevation of every geophone and every shot point** with a level and staff, a total station or a GNSS/GPS receiver. Write them on the field sheet.
3. Enter them as **Z** in the recording software if it allows it, so they are saved in every file.
4. Otherwise, in Refraction Studio open **Geometry → Ground profile** and paste or import the surveyed points (position and elevation) as CSV. Geophone and shot elevations are interpolated from this profile.

9. Before you leave the site

- Copy the data folders to a USB stick **and** keep them on the laptop: two copies.
- Check the field sheet: every file has a shot position, the spacing, the channel 1 end, the wave type and notes.
- Photograph the line from both ends with the geophones in place.
- Note the GNSS/GPS coordinates of both ends of the line.
- Count the geophones before you pack them.

10. Install Refraction Studio and analyse the data

INSTALL

1. Run **RefractionStudio-1.0.0-win-x64-setup.exe** on a computer running Windows 10 (version 1809 or later) or Windows 11, 64-bit.
2. Windows may show **Windows protected your PC** because the installer is not yet code-signed: choose **More info → Run anyway**.
3. Follow the installer and start Refraction Studio. It is the full version: there is no licence key, no activation and no internet connection needed.
4. Next time, start it from **Start → Refraction Studio** or the desktop shortcut.

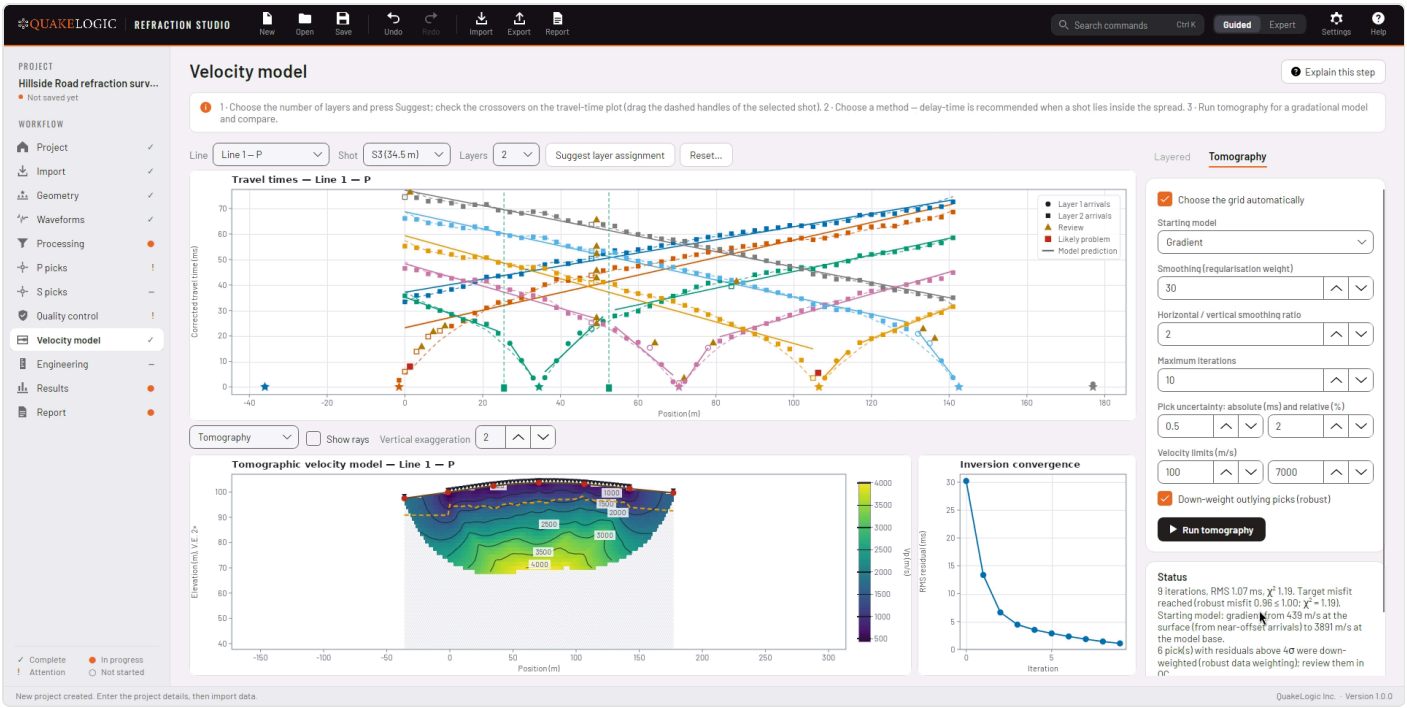
Choose **Explore the sample** on the start page once: it is a complete P and SH survey that shows every step below.

ANALYSE YOUR SURVEY

The workflow on the left of the window lists the steps in order, with the status of each one.

1. **Project.** Enter the project, client and site; they appear on the report.
2. **Import.** Choose the files or the folder of one line. Check the validation findings, choose the wave type (P or SH) and, for SH, **Opposite SH impacts**, then import.
3. **Geometry.** Compare the positions with the field sheet. Set the role of the reference geophone, if you used one, and enter the ground profile.

4. **Waveforms and Processing.** Look through the shots. Remove the DC offset and, if surface waves hide the first breaks, add a causal high-pass filter.
5. **P picks and S picks.** Press **Pick automatically**, accept the high-confidence picks and review the rest: click a trace to place or move a pick.
6. **Quality control.** Work through the findings; each one says what to check.
7. **Velocity model.** Press **Suggest layer assignment**, check the crossovers, apply the delay-time method, then run tomography for a gradational model.
8. **Engineering.** Vs30, average velocities, Poisson's ratio and the elastic moduli.
9. **Results and Report.** Export the tables and figures, then generate the PDF report. Save the project with **Ctrl+S**.



Velocity model page: travel times, the layered model and tomography

The user guide (in the software under **Help**, F1, and as a PDF in this package) explains every setting. The training video shows the whole workflow in about six minutes.

11. Field checklists and field sheet

BEFORE YOU LEAVE THE OFFICE

- Seismograph, laptop with recording software, cables, charged batteries
- Line cable and spare connectors
- Vertical geophones (P) and horizontal geophones (SH), one per channel plus two spares
- Trigger switch or trigger geophone; optional reference geophone
- Sledgehammer and strike plate; shear beam and its load (vehicle or weights)
- Measuring tape, flags, pegs, small hammer
- Level and staff, total station or GNSS receiver

- Printed field sheets, pencil, safety equipment
- Recording and trigger tested in the office

ON SITE, BEFORE THE FIRST SHOT

- Line straight, geophone and shot positions marked and written down
- Channel 1 end decided and noted
- P geophones upright and firm; SH geophones level, across the line, all turned the same way
- Every channel found by the recording software; trigger tested
- Settings: 0.125–0.25 ms (P) or 0.25–0.5 ms (SH) sampling, record length for the farthest offset, no field filters, the same gain on every channel
- Noise checked: no dead or noisy channels

FIELD SHEET

File	Line / wave	Shot position (m)	Hits	Channel 1 at (m)	Spacing (m)	Notes (noise, weather, elevations)

12. Troubleshooting

Problem	Likely cause	What to do
A channel is missing in the recording software	Loose connector or faulty take-out	Re-seat the connectors from channel 1 outwards
Nothing records when the hammer hits	Trigger cable loose, or trigger geophone too far from the plate	Re-plug the trigger; put the trigger geophone right next to the plate
Recording starts by itself	Trigger disturbed by footsteps or wind	Keep people away from the trigger geophone; raise the trigger threshold
One trace is flat	Geophone unplugged or broken	Re-plug or swap the geophone
First breaks are lost at the far end	Not enough energy	Stack more hits, use a heavier source, shorten the spread or add off-end shots
First breaks jump earlier or later on a whole shot	Trigger timing error	Use a reference geophone; correct the shot time in Processing → Shot time correction
L and R SH records look the same	Geophones not turned the same way, or the beam hit on the same side	Check the geophone orientation and the hit direction; re-record
Refraction Studio warns that the sample rates differ	Settings changed between shots	Keep one sample interval per line; import different rates as separate lines
The layered model is "not available"	Too few picks assigned to a refractor	Press Suggest layer assignment, then check the crossovers on the travel-time plot
Vs30 is not available	The SH model does not reach 30 m (100 ft)	Longer SH spread or off-end shots; extrapolation is possible but is labelled

Help: email support@quakelogic.net or call QuakeLogic on +1 (916) 899-0391. Email response time: 3 business days.